IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

POWER SYSTEM ECONOMICS

Tuesday 2 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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The Questions

Question 1

(a) Describe the significance of the concept of system marginal price and explain why it varies with time in the case of electricity.

[2]

(b) Three generators are available to supply a two-period demand of $D_1 = 200\ [MW]$ and $D_2 = 800\ [MW]$. The cost of generating power $C_{i,t}(P_{i,t})$ corresponds to each generator i at hour t is:

$$C_{1,t}(P_{1,t}) = 700 + 1 * P_{1,t} + 0.05 * P_{1,t}^2\ [\frac{£}{h}]$$

$$C_{2,t}(P_{2,t}) = 600 + 6 * P_{2,t} + 0.25 * P_{2,t}^2\ [\frac{£}{h}]$$

$$C_{3,t}(P_{3,t}) = 500 + 15 * P_{3,t} + 0.55 * P_{3,t}^2\ [\frac{£}{h}]$$

Calculate:
- the optimal production of each generator at each hour,
- the system marginal cost at each hour,
- the total cost of operating the system,
- the total profit of each generator.

[8]

(c) Modify the dispatch found in (b) that respects the constraints on the minimum and maximum outputs of each of the generators:

$$P_1^{min} = 100\ [MW] \text{ and } P_1^{max} = 800\ [MW]$$
$$P_2^{min} = 20\ [MW] \text{ and } P_2^{max} = 110\ [MW]$$
$$P_3^{min} = 20\ [MW] \text{ and } P_3^{max} = 80\ [MW]$$

Calculate:
- the optimal production of each generator at each hour,
- the system marginal cost at each hour,
- the total cost of operating the system,
- the total profit of each generator.

[8]

(d) The total profit of generator 3 in (c) is negative. Explain why this is the case. If the generators are free to participate in the market based on their profitability, discuss how this generator can avoid the economic losses.

[2]

Question 2

(a) Explain briefly why electricity markets may lend themselves to be imperfect competitive markets.

[4]

(b) Consider a market for electrical energy that is supplied by two generating companies whose cost functions are:

$$C_A = 75 * P_A \left[\frac{£}{h}\right]$$

$$C_B = 62 * P_B \left[\frac{£}{h}\right]$$

Where P_A and P_B are production levels of the two generating companies respectively. The inverse demand function for this market is estimated to be:

$$\pi = 250 - 2D \left[\frac{£}{MWh}\right]$$

Where π and D are the electricity price and demand respectively.

(i) If we assume a Bertrand model for the competition in this market, calculate the production of each of the companies, the resulting market price and profits made by both companies.

[2]

(ii) If we assume a Cournot model of competition, form a table to identify the Nash Equilibria (NE) of this market, i.e., market price, demand quantity and profit of each company for different levels of production for each of the companies. Consider a range of production levels for company A of 20MW, 25MW, 30MW and 35MW, and for company B of 25MW, 30MW, 35MW and 40MW.

[8]

(c) It is observed in (b)-(ii) that one of the cells of the derived Cournot table involves higher profits, for both competing companies, than the one corresponding to the NE. Explain why this cell does not express the equilibrium state of the market.

[2]

(d) State the optimality conditions and evaluate the exact NE of this market. At the NE, calculate the market price, demand quantity and profit of each company.

[4]

Question 3

(a) For the system shown in Figure Q3, and the data given in Tables Q3-1 and Q3-2, calculate the unconstrained economic dispatch as well as the resulting nodal prices and power flows in each line.

[5]

(b) Demonstrate that line 1-3 is overloaded if the capacity limits of the lines are considered and the dispatch calculated in (b) is implemented. Show how this overload can be eliminated by:

(i) increasing the output of generator A and reducing the output of generator C.

(ii) increasing the output of generator B and reducing the output of generator C.

Calculate the hourly cost of options (i) and (ii) above. Which of the two options is preferable and why?

[10]

(c) Calculate the nodal prices when the more preferable of the two options compared in (b) is implemented.

[5]

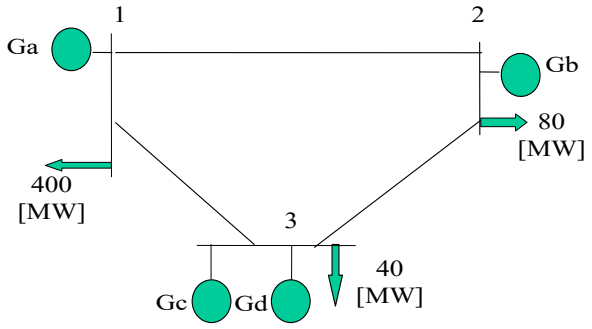


Figure Q3: System layout.

Table Q3-1 Generator data.

Generator	Capacity (MW)	Marginal cost (£/MWh)
Ga	200	15
Gb	150	12
Gc	150	10
Gd	400	8

Table Q3-2 Line data.

Line	Per unit reactance	Capacity (MW)
1-2	0.2	250
1-3	0.3	250
2-3	0.3	250

Question 4

(a) Two areas, A and B, of the Borduria power system is linked by a transmission line operated by the Borduria Transco. In winter, the demand in area A is 1,000MW while the demand in area B is 2,500MW. In summer, the demand in area A is 500MW while the demand in area B is 1,500MW. The cost of generation in the areas can be modelled by the following expressions:

$$C(P_A) = 1 \cdot P_A + 0.001 \cdot P_A^2 \quad [£/h] \text{ for area A}$$

$$C(P_B) = 2 \cdot P_B + 0.002 \cdot P_B^2 \quad [£/h] \text{ for area B}$$

Determine the electricity prices and optimal levels of generation in areas A and B for each of the seasons,

(i) when there is no transmission link between areas A and B; and

[2]

(ii) when areas A and B are connected with large network capacity (imposing no network constraints).

[2]

(iii) What are the benefits and losses to consumers and generators in Areas A and B, when comparing cases (i) and (ii)?

[2]

(b) If necessary, modify the levels of generation computed in (a)-(ii) to take into consideration the capacity of the transmission link at 800MW. What are the electricity prices in areas A and B in each of the seasons? Calculate the hourly generation cost in areas A and B for each of the seasons.

[4]

(c) Continuing (b), assuming that the duration of the winter and summer period is 2,700h and 6,060h respectively, calculate:

(i) the total annual generation costs

[2]

(ii) the total annual revenue received by the generators and the charge imposed on demand if electricity is priced at locational short-run marginal prices.

[2]

(iii) the total annual transmission revenue received by Borduria Transco in this case.

[2]

(d) Borduria Transco has proposed to the Borduria Regulator to reinforce the line between the two areas by 900 MW. Assuming that the annuitised investment cost of the additional line is 5,000,000 £/year, determine if this proposed investment is justified.

[4]